CEGE0024

Structures and Materials: Plastic Analysis and Finite Element Modelling

Dr. Leo de Waal
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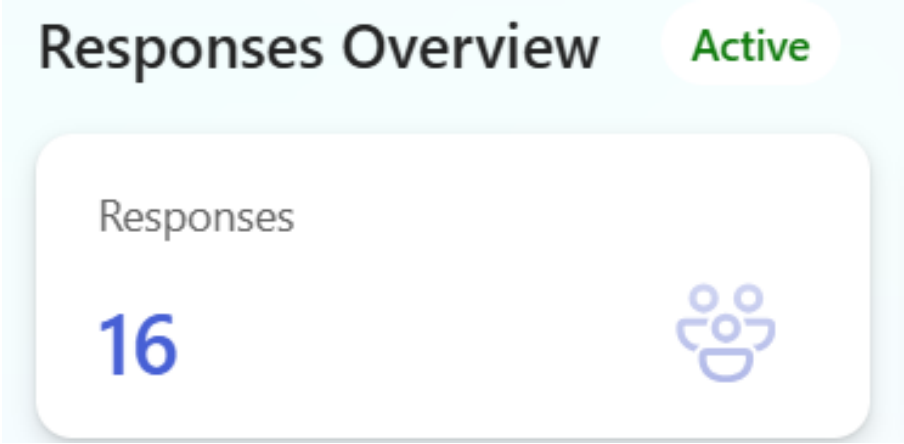
Feedback



Feedback

What should I continue doing because it's working well?

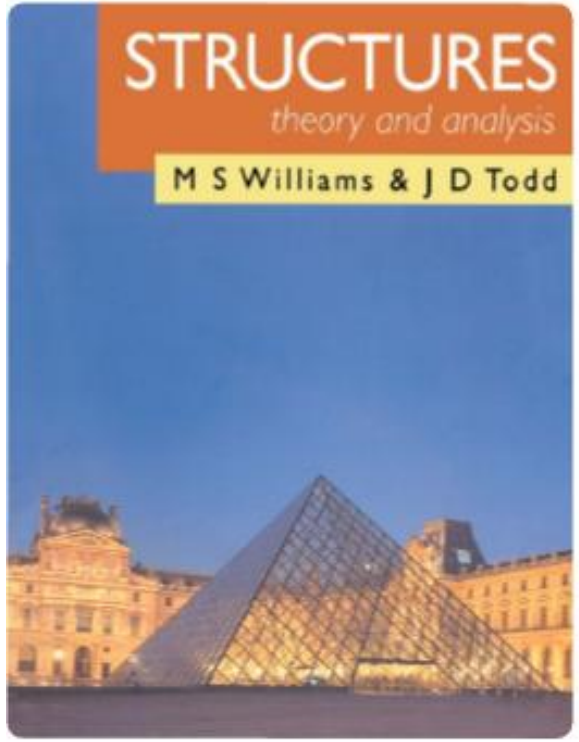
- Going through examples
- Making it easy to ask questions

What should I start doing in the course to improve your leaning?

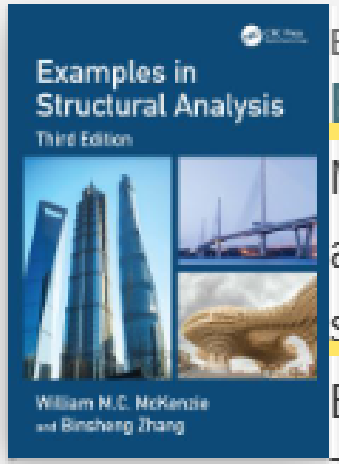
- Slow down, went a bit too fast
- Vidcast to enable review of lecture
- Releasing PowerPoints ahead of the classes
- More examples, exam practise questions

Feedback

- Example problems



Structures: Theory and Analysis ...
Williams, Martin



BOOK

Examples in structural analysis.

McKenzie, W. M. C. (William M. C.), author. Zhang, Binsheng, Dr. author. McKenzie, W. M. C. (William M. C.). Examples in structural analysis.

Boca Raton : CRC Press, 2022

Third edition / William M.C. McKenzie, Binsheng Zhang.

View Online

Feedback

What should I stop continue doing because it's working well?

- Going too fast, especially lectures 4-5
- Mistakes within worked examples

What have you found to be the hardest concept covered so far?

- Virtual work
- Continuous beams
- Portal frame
- ICR
- Mp for frames
- Lectures 4-5
- Finite element

Feedback

How satisfied are you with your learning so far and why?

- Quite satisfied – going through examples
- Decently satisfied
- Okay, could do with more resources to supplement
- I didn’t understand too much
- Content too difficult

Sixth session objectives

Brief introduction to FE modelling and understand the effect of the following **fundamental aspects** for applying an FE model:

- Discretisation
- Shape functions
- Element types

FE Modelling

"*One of the big problems facing engineers is that we can solve only the simplest of structures by hand because the mathematical process rapidly becomes too complex. This means that we need techniques such as FEA working on computers to solve them for us. Our job is to produce* ***good answers****, and we do that by putting* ***good data*** *into appropriate software. But how do we know what is good data and good software?"* [2]

Learning outcomes:

- **Understand** the importance of **implementing an effective structural Finite Element modelling** process and its **fundamental principles**

This is not a fundamental FE theory course.

However, you need some understanding of the theory to appreciate the importance of your modelling choices.

[2] Debney, P., (2020).

Reference material

BOOK

Modern Structural Analysis - Modelling Process and Guidance

MacLeod, Iain A. (Iain Alasdair)

London: ICE Publishing, 2005

❝ This book is essential reading for 21st century practitioners and students who need to do structural analysis...❞

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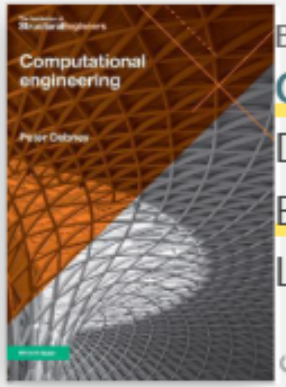

BOOK

Computational engineering / Peter Debney.

Debney, Peter, author. Institution of Structural Engineers (Great Britain)

London : Institution of Structural Engineers, 2020

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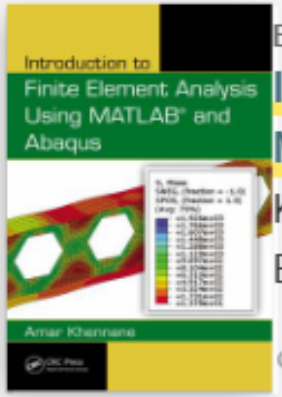


BOOK

Introduction to finite element analysis using MATLAB and Abaqus / Amar Khennane.

Khennane, Amar, author.

Boca Raton : Taylor & Francis, 2013

View Online ›

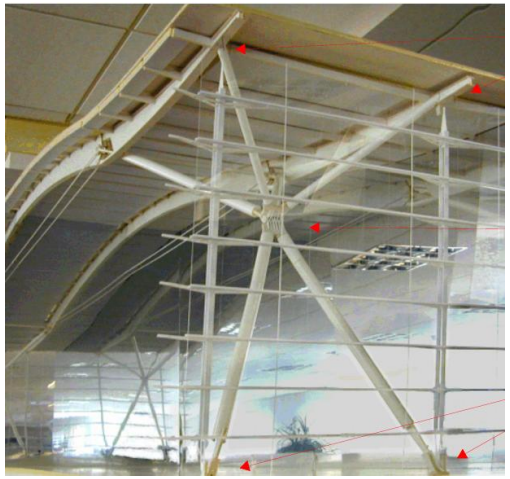
What is a FE model?

- FE method is a numerical technique that can be used to obtain an **approximate solutions** to engineering problem

↓

They are wrong but can be useful if done correctly

Real system



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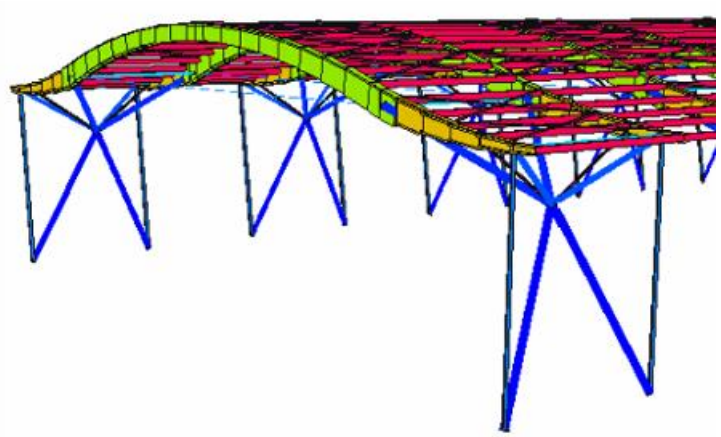
Mathematical representation



The FE modelling process at a glance

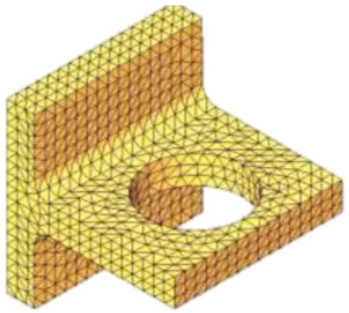
Real structure

Idealisation

Mathematical model

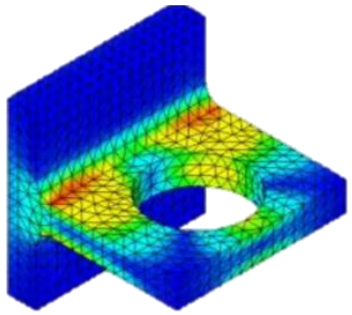
- Analysis type
- Constitutive relationship (i.e. Material properties)
- Boundary conditions (Restraints and loads)

Discretisation



- Element type
- Element order

Numerical Solution



FE Solvers are very powerful and they will generally give you a solution

Garbage in = Garbage out

How do we know if we have implemented and effective FE model?

*To answer this question effectively - we first need some basic understanding of how FE models work – **some fundamental principals***

[5] Sharma, K. (2019).

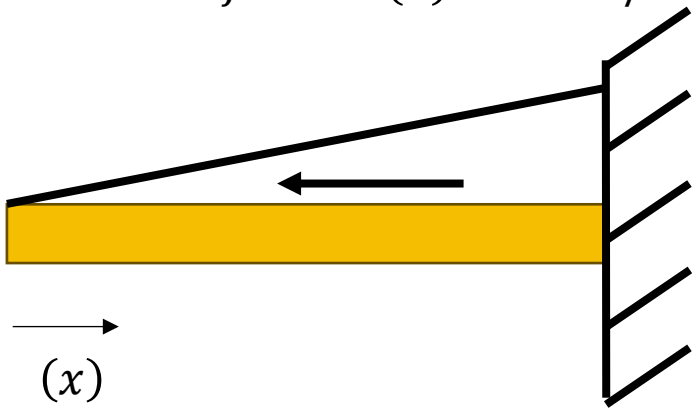
How it works

- FE model solves a boundary value problem – solves differential equations (PDEs) within a specific domain

- Typical ingredients in the context of physical systems:
    1. Domain with a boundary
    2. Kinematics: a field of interest what are we solving for i.e. displacement, velocities, temperature etc.
    3. Equilibrium/balance principles: laws of physics (i.e. force balance) - Equation we want to solve
    4. Constitutive equations: How the kinematics relates to the equilibrium (material response)
    5. Boundary conditions

How it works: Exact solution

$Traction\ force\ T(x) = -x\ N/mm$



$Length = 1000mm$
$Area = 500mm^2$
$E = 2 * 10^5 MPa$

- Kinematics: We want to find the displacement field $u(x)$ along the bar.

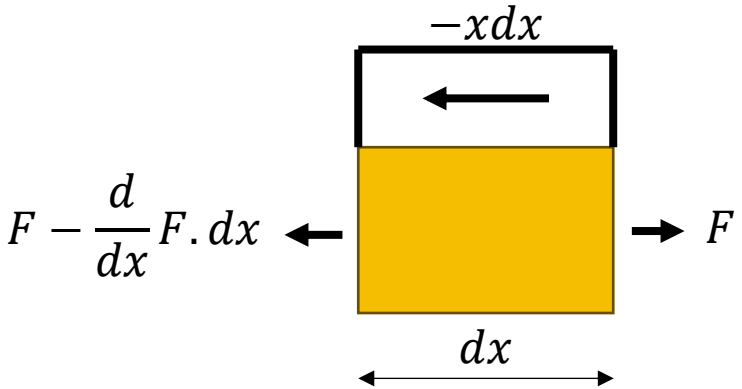
$$\varepsilon = \frac{d}{dx}u(x)$$

- Equilibrium:

$$F - \left(F - \frac{d}{dx}Fdx\right) - xdx = 0$$

$$\frac{d}{dx}F - x = 0$$

$$\frac{d}{dx}\sigma A - x = 0$$



- Constitutive equation:

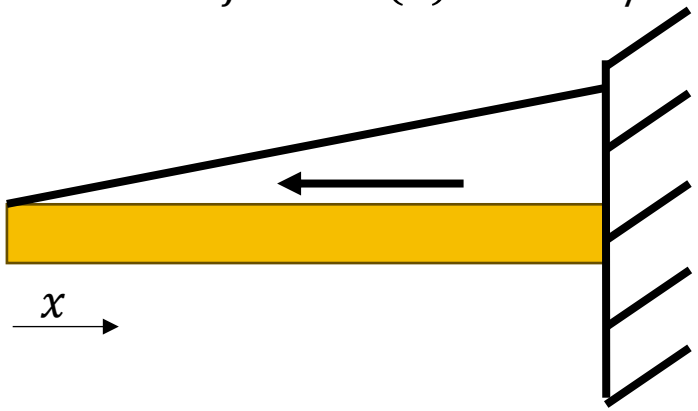
$$\sigma = E\varepsilon$$

$$EA\frac{d}{dx}\varepsilon - x = 0 \rightarrow EA\frac{d^2u}{dx^2} - x = 0$$

How it works: Exact solution

$$EA\frac{d^2u}{dx^2} - x = 0$$

$Traction\ force\ T(x) = -x\ N/mm$

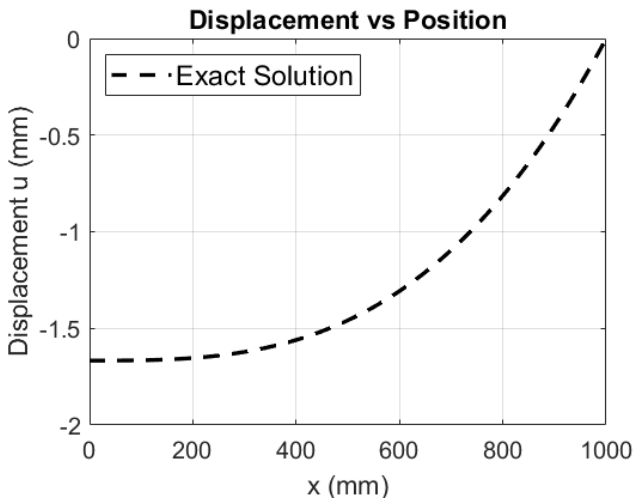


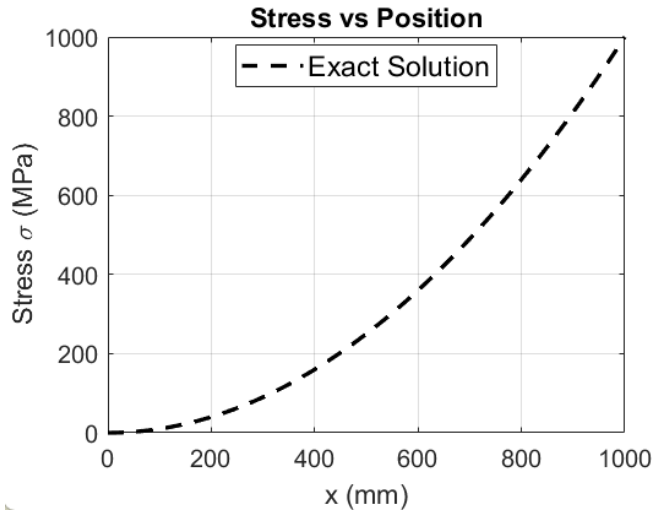
$Length = 1000mm$
$Area = 500mm^2$
$E = 2 * 10^5 MPa$

- Boundary conditions:

$\sigma(x = 0) = 0$
$u(x = 1000mm) = 0$

- This PDE can be solved analytically quite easily:
  - Integrate once to give the stress field and apply BC
  - Integrate twice for the displacement field and apply BC





How it works: FE

For complex geometries, it is difficult to solve for an analytical expression for the field $u(x)$, so instead we want to discretize and just solve at specific nodal locations $\{r\}$.

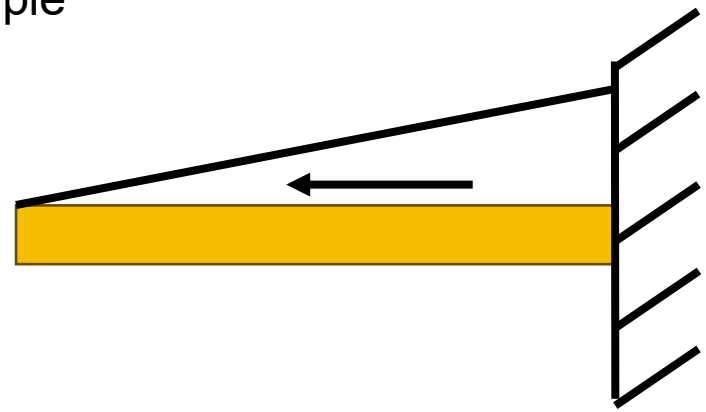
$$u(x) = [N]\{r\}$$

Where: $[N]$ is the shape function that is used to approximate the displacement field within the element using nodal degrees of freedom $\{r\}$.

Here there are three key modelling choices for FEM:

1. What element type can sufficiently model the system

Example



Modelling choice: element type

- Here we are trying to solve for axial displacements and stress due to the axially applied loads. The only degree of freedom that are therefore of interest are axial degrees of freedom.
- Therefore, we will choose a truss element (*the element options will be discussed later in the lecture*)

How it works: FE

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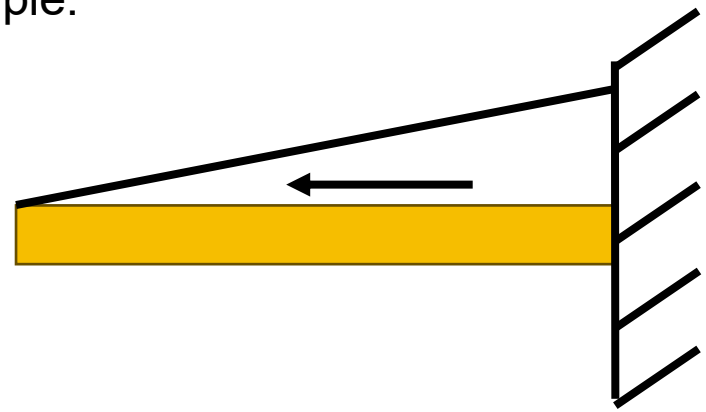
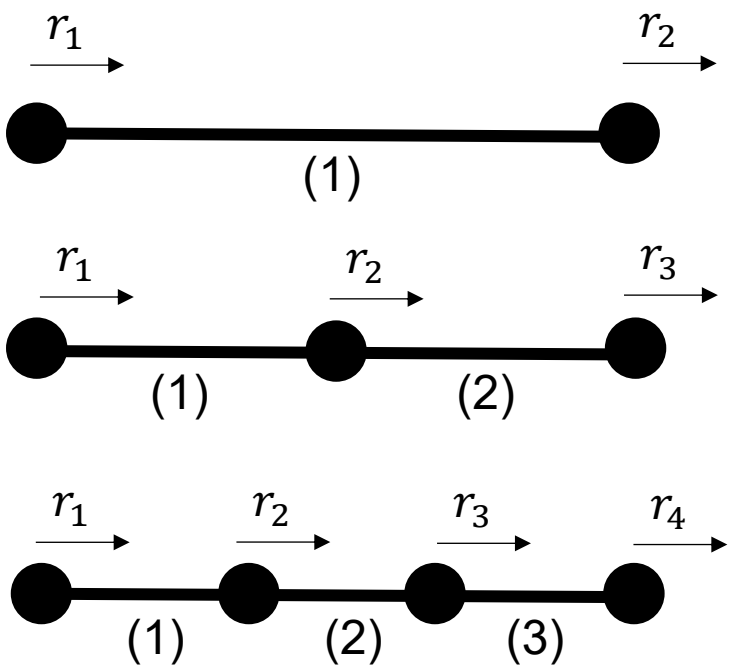
Here there are three key modelling choices for FEM:

1. What degrees of freedom are important
2. **How you choose to discretize (i.e. how many elements)**

Example:



Modelling choice: Level of discretization



How it works: FE

For complex geometries, it is difficult to solve for an analytical expression for the field $u(x)$, so instead we want to discretize and just solve at specific nodal locations $\{r\}$.

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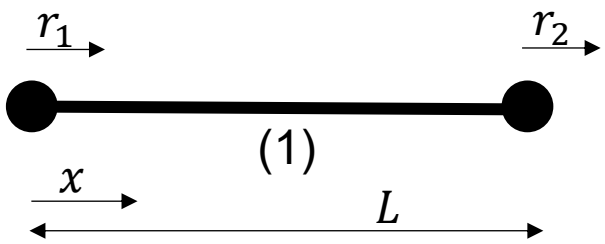
Where: $[N]$ is the shape function that is used to approximate the displacement field within the element using nodal degrees of freedom $\{r\}$.

Here there are three key modelling choices for FEM:

1. What degrees of freedom are important
2. How you choose to discretize (i.e. how many elements)
3. **How you wish to approximate your displacement field.**

Truss element example:

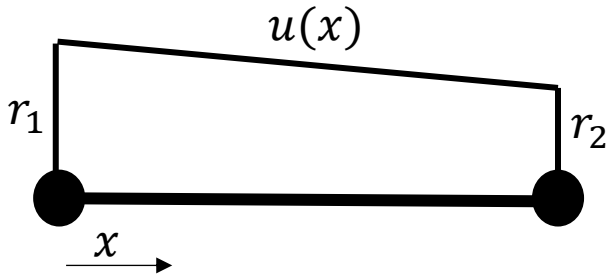
$$u(x) = [\boldsymbol{N}]\{r\}$$

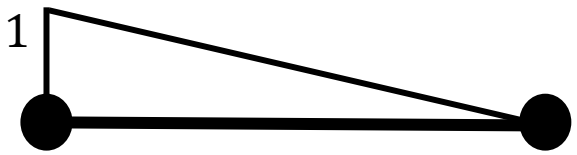


$$u(x) = [N_1 \quad N_2]\begin{Bmatrix} r_1 \\ r_2 \end{Bmatrix}$$

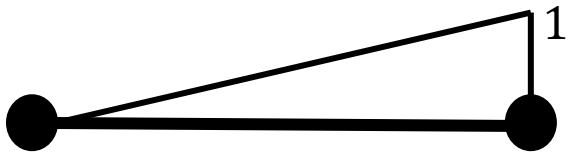
$$u(x) = N_1 r_1 + N_2 r_2$$

Modelling choice: Assume a linear displacement field exists within the element





$$N_1 = 1 - \frac{x}{L}$$



$$N_2 = \frac{x}{L}$$

How it works: FE

Using this approximation, the FEM method transforms the equilibrium equation into a system of simultaneous algebraic equations. For **linear static analysis**, these simultaneous algebraic equations have the form:

$$\{F\} = [K]\{r\}$$

Where: $[K]$ is the stiffness matrix and is ***found numerically*** *(it* depends on the choice of shape function.)

NOTE: Direct stiffness method also discretises the system, however the stiffness matrix is found ***analytically*** and the ***displacements are discontinuous***.

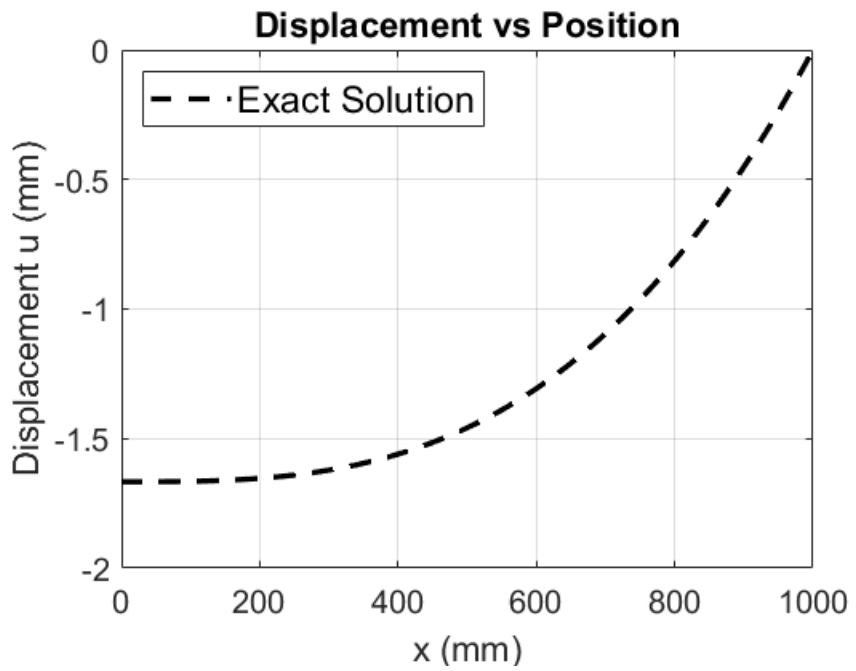
Applying the boundary conditions, you can solve for the nodal displacement $\{r\}$ by inverting $[K]$.

$$\{r\} = [K]^{-1}\{F\}$$

*Note: to be able to invert the stiffness matrix the $\mathbf{det}[\boldsymbol{K}] \neq \mathbf{0}$. If the **matrix is singular** your FE model **will not be able to solve**. Look at your constraints which you applied or if the stiffness values are very small.

How it works: FE

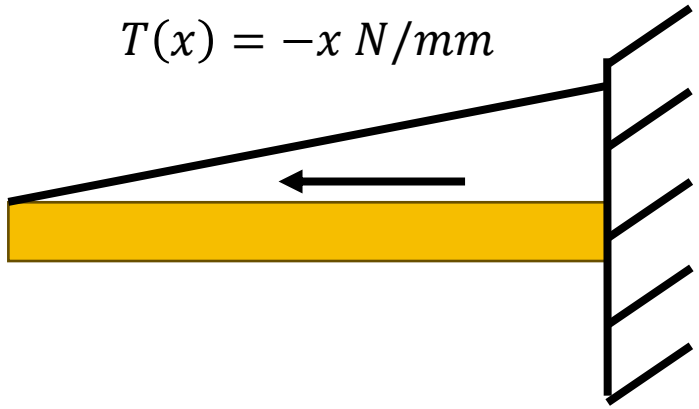
Based on these modelling choices what will the solution look like?



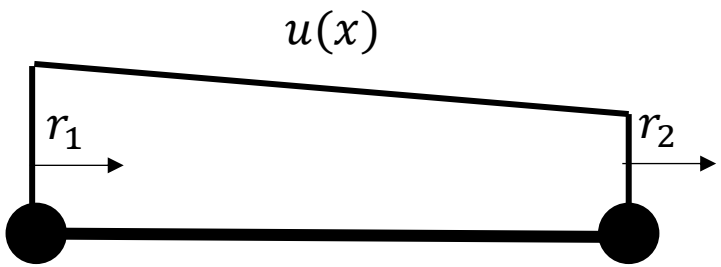
Truss element example:

$$T(x) = -x\ N/mm$$

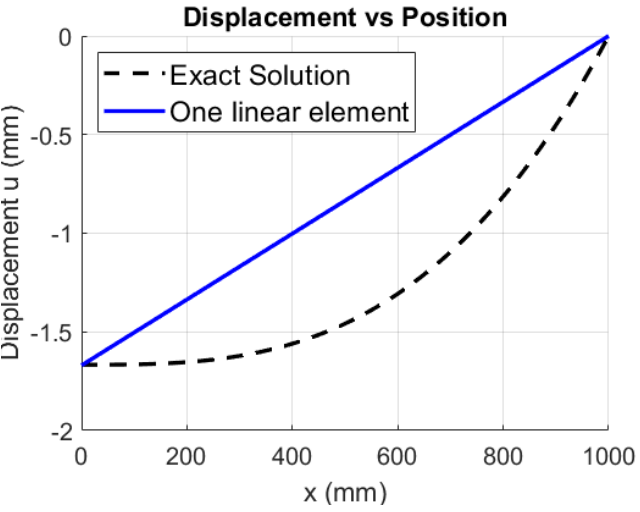
$$Length = 1000mm$$
$$Area = 500mm^2$$
$$E = 2 * 10^5 MPa$$

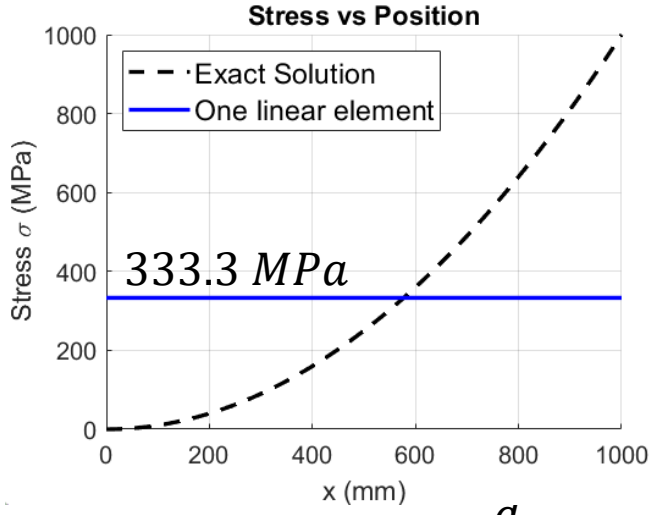


Modelling choice: One element+ linear shape function



How it works: FE





$$\sigma = E\varepsilon = E\frac{d}{dx}u(x)$$

Truss element example:

$$T(x) = -x\ N/mm$$

$$Length = 1000mm$$
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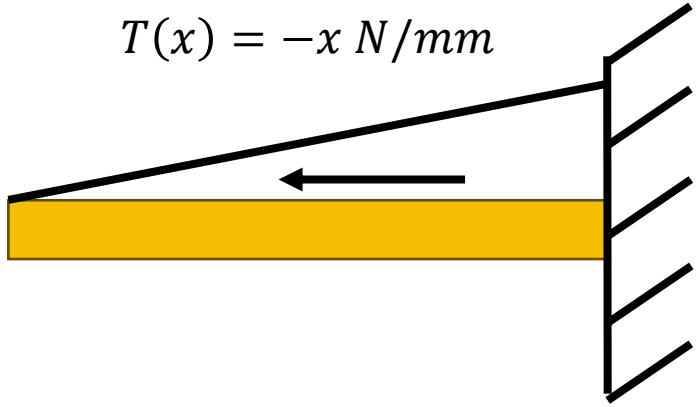
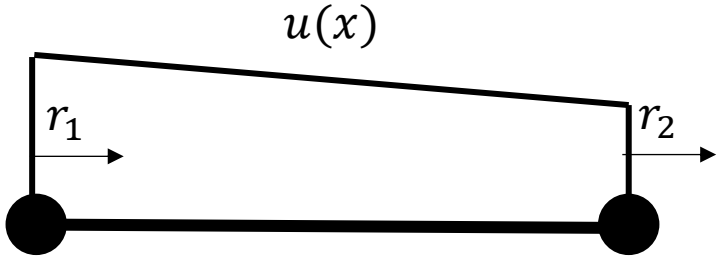
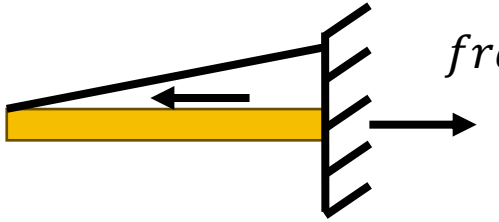


Modelling choice: One element+ linear shape function



- External equilibrium

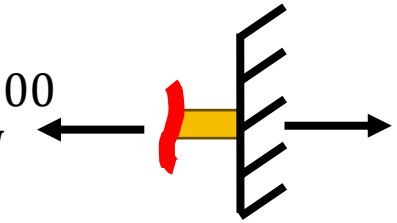


$$from\ \{F\} = [K]\{r\}$$
$$f_2 = 500kN$$

$$\frac{1000*1000}{2} = 500kN$$ satisfied

- Internal equilibrium

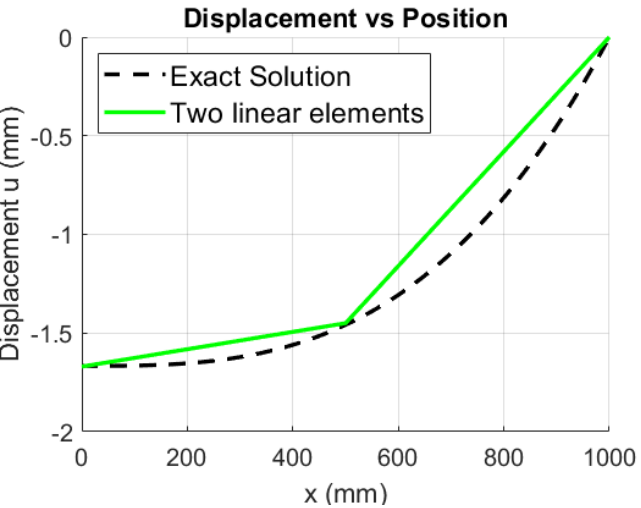
$$333.3 * 500 = 167kN$$

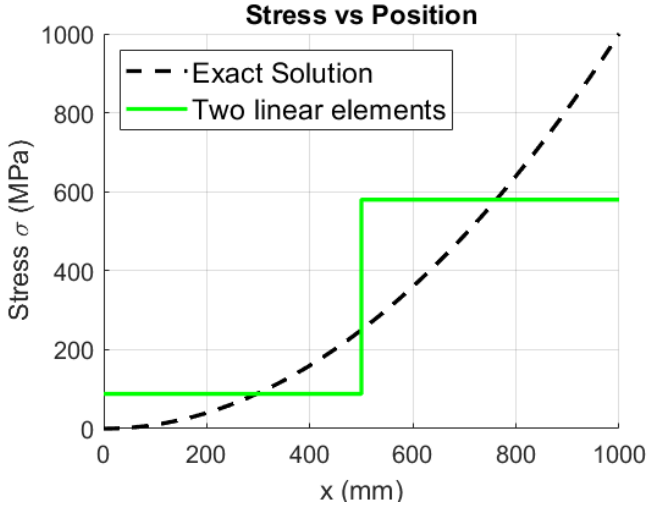


$$f_2 = 500kN$$

$$167kN = 500kN$$ Not satisfied (It is approximate!)

How it works: FE





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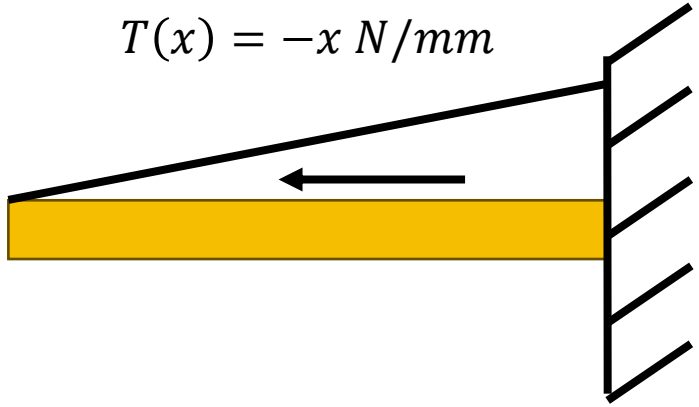
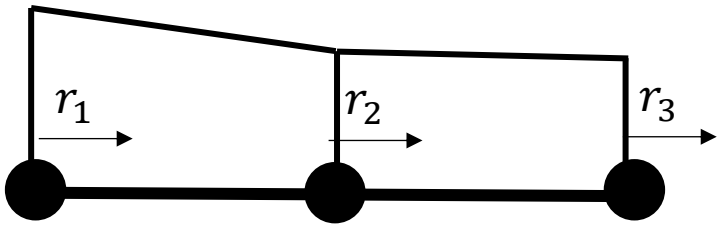


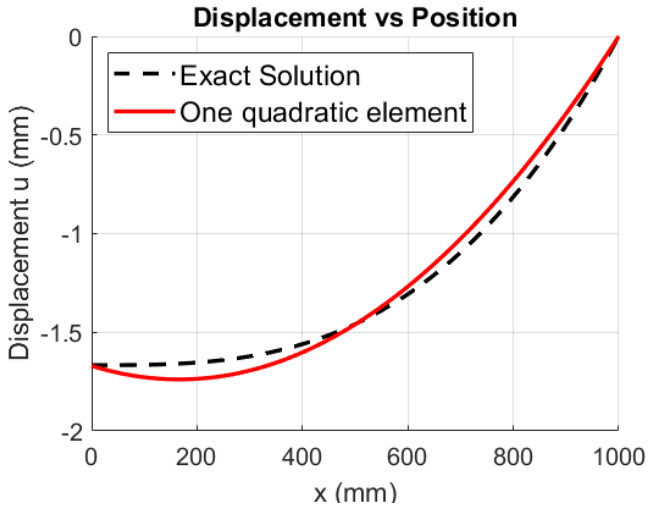
Modelling choice: Two elements + linear shape functions

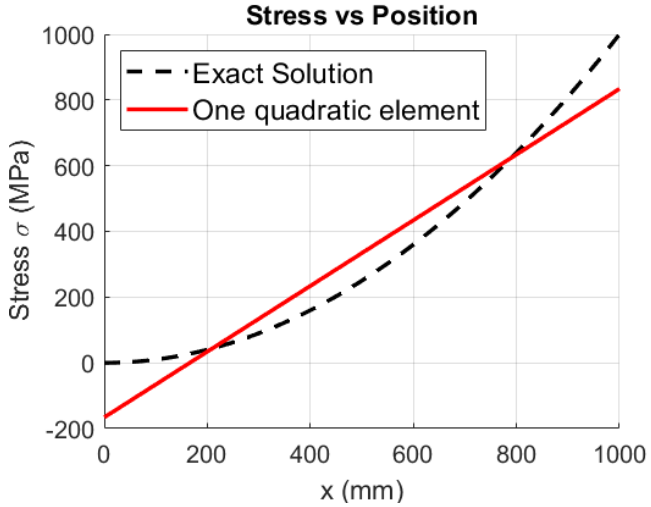
$$u(x)$$



- External equilibrium
  Satisfied
- Internal equilibrium
  Not satisfied (It is approximate!)

How it works: FE





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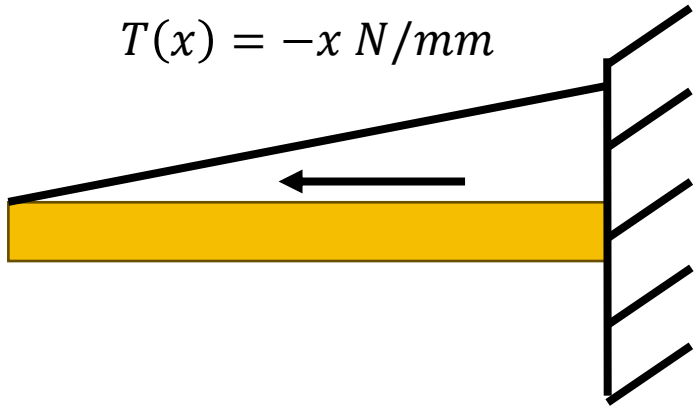
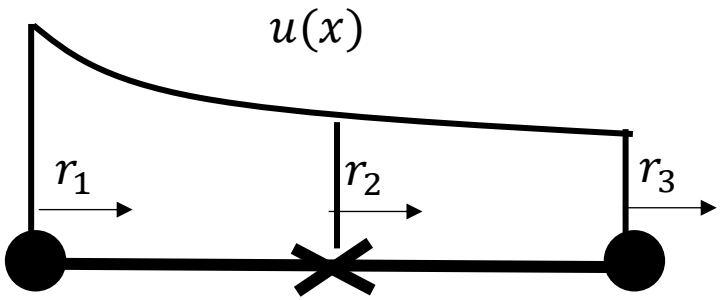


Modelling choice: One elements + quadratic shape function

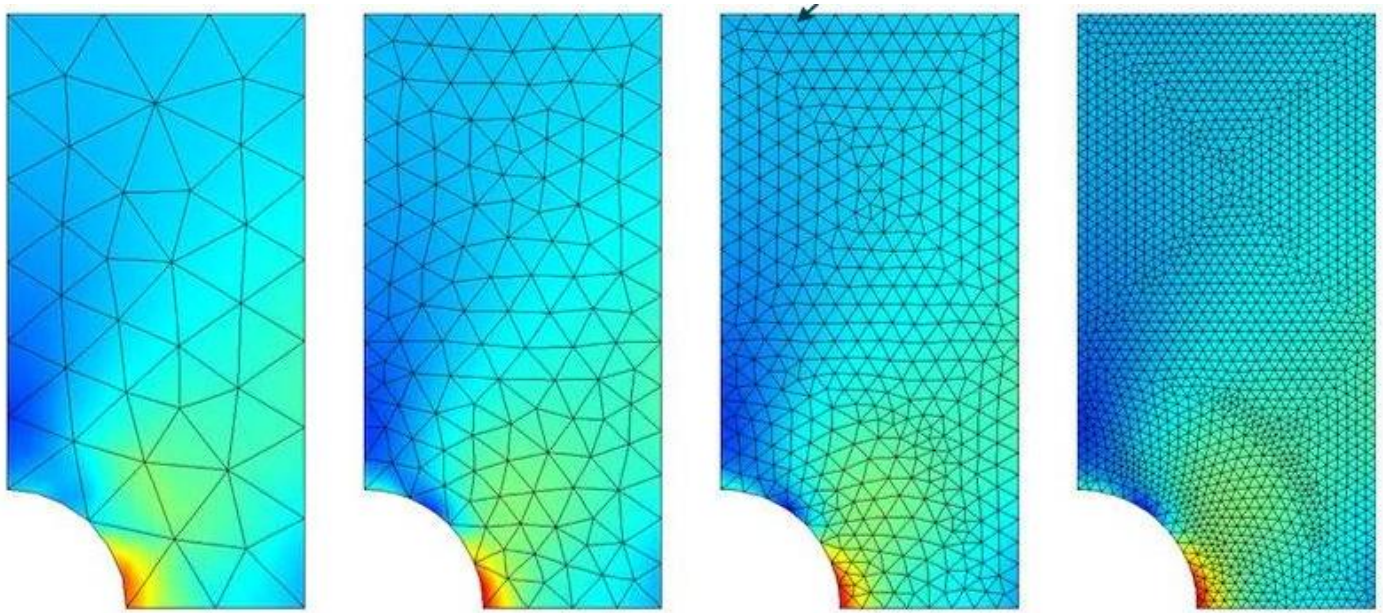


- External equilibrium
  Satisfied
- Internal equilibrium
  Not satisfied (It is approximate!)

Note: In this example the displacement field is a cubic function, so to recover the exact solution we would need to assume a cubic shape function for the element

General points on discretisation

Coarse mesh → Fine mesh

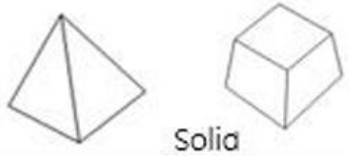


We will discuss mesh sensitivity study in the next lecture!

- A finer mesh density → generally increases accuracy
- Computationally → costs more
- Higher order shape functions → require less discretisation, and generally have less numerical issues

[6] Duffour, P (2026).

Element types

Type	Actual Models	Finite Element Expressions (Geometric Properties Defined by Nodes)
1D	Rod (Truss) Beam	Length (L)
2D	Shell, Plane Stress, Plane Strain, Axisymmetric, etc	Area (A)
3D	 Solid	volume (V)
Misc.	Spring, Mass, Rigid Link, etc.	-

Reducing the dimension of the problem through assumptions

[5] Sharma, K. (2019).

Element types

<u>1D element</u>

- Member is slender: length is much larger than the cross-section dimensions → the members behavior can be represented along its centerline; field variable varies in one-dimension only.
- The stress variation within the cross-section is not explicitly solved
- Cross-section properties have to be defined for all elements.

Element types

<u>1D element:</u> Truss element

- Kinematic assumption: only axial deformation
- No rotational degrees of freedom, can only carry axial loads

In 2D with first order shape function:

$$\{F\} = [K]\{u\}$$

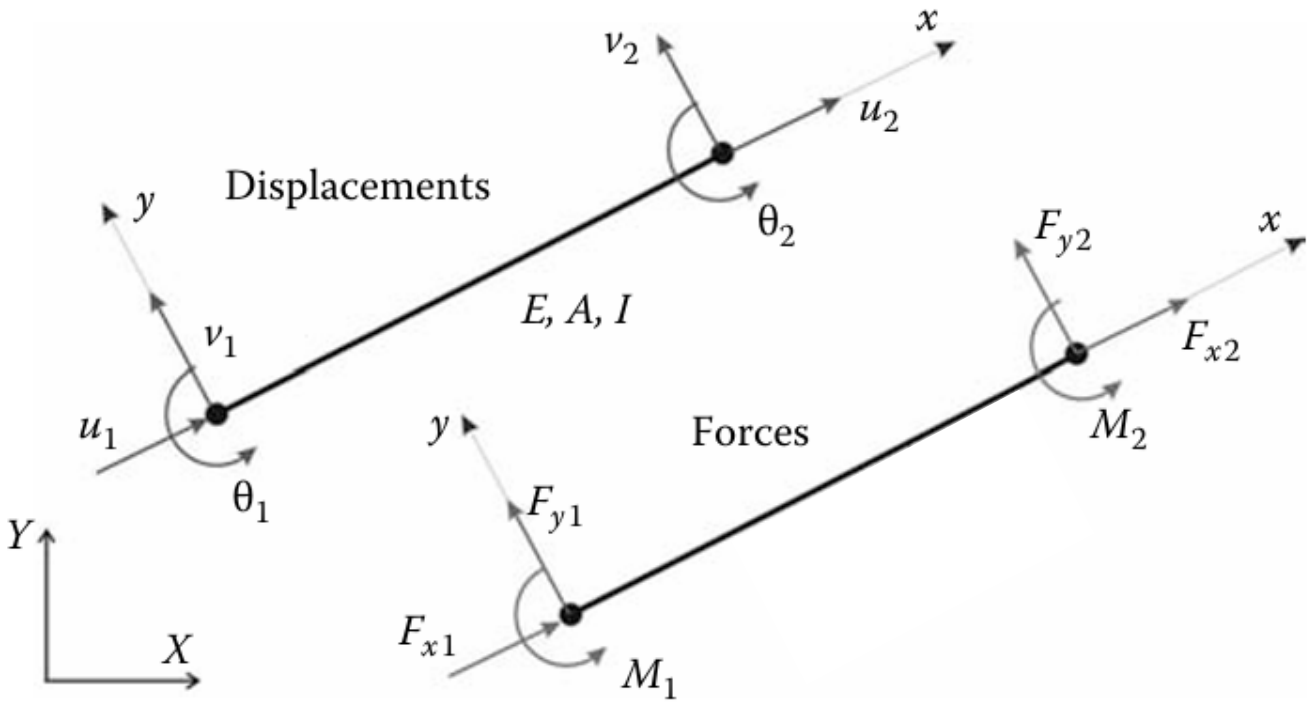
$$\begin{Bmatrix} F_{x1} \\ F_{y1} \\ F_{x2} \\ F_{y2} \end{Bmatrix} = [K] \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix}$$

[3] Khennane, A. (2013).

Element types

1D element: Beam-column elements

- Kinematic assumption: Axial deformation and beam theory (e.g. Euler-Bernoulli beam theory, Timoshenko beam theory etc.)



In 2D with first order shape function:

$$\{F\} = [K]\{u\}$$

$$\begin{Bmatrix} F_{x1} \\ F_{y1} \\ M_1 \\ F_{y1} \\ F_{x2} \\ M_2 \end{Bmatrix} = [K] \begin{Bmatrix} u_1 \\ v1 \\ \theta_1 \\ u_2 \\ v_2 \\ M_2 \end{Bmatrix}$$

[3] Khennane, A. (2013).

Element types

Let's have a look at a simple frame example…

Beam elements or truss elements?

[6] Duffour, P (2026).

Element types

Bar elements

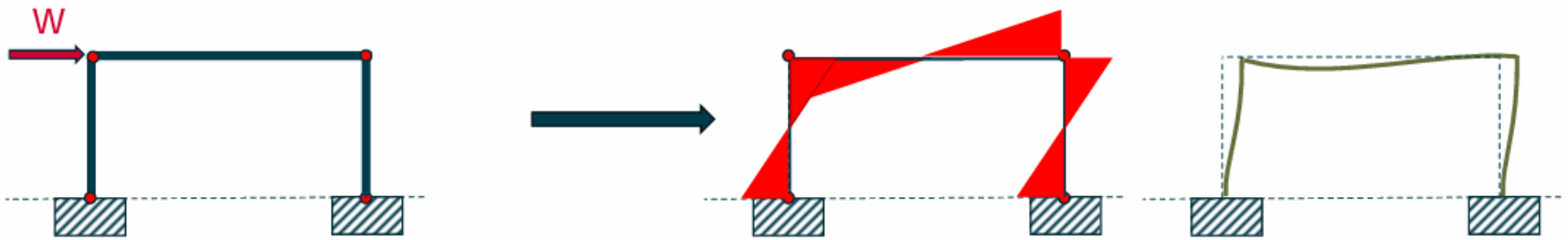
Bar elements can only model axial forces

Unstable

Since bar elements can only capture axial forces, this frame is effectively pin jointed and therefore unstable

$[K]$ **is singular!**

Beam elements



Beam elements can model axial + bending + shear

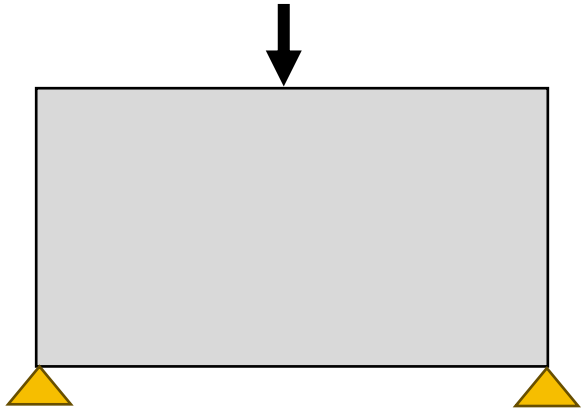
The elements transfer the forces they can model. So here the frame is moment-resisting and can resist the load.

[6] Duffour, P (2026).

Element types

Can beam-column elements be used to model:

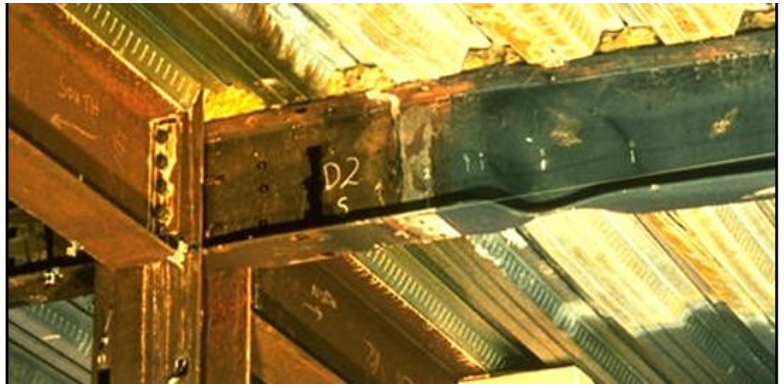
- Thick short beams?



- The thermal induced buckling of railway tracks?

[7] Mandal and Lees (2016).

- The thermal induced local buckling



[4] Newman et al. (2000)

Element types

<u>2D element</u>

- One dimension (thickness) is much smaller than the other two dimensions →out-of-plane stresses are negligible; field variable varies in two dimensions. Shell geometry is defined at the mid-surface.

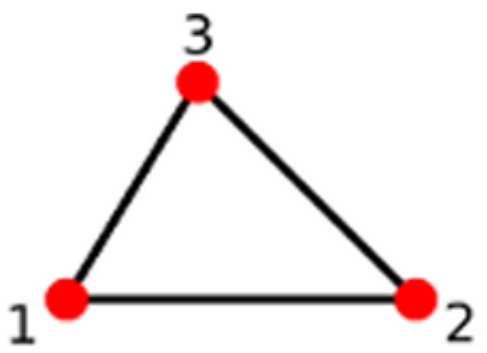
- Thickness has to be defined for the element.

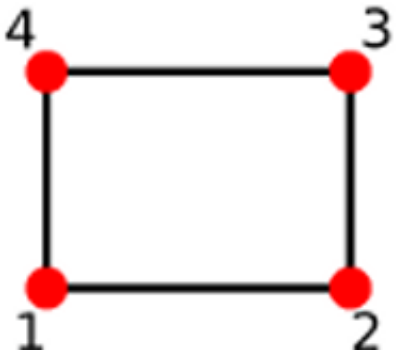
Element types

<u>2D element:</u>

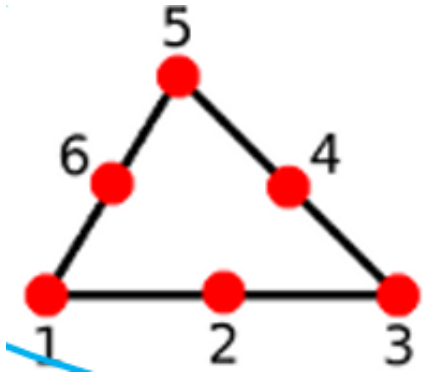
- Displacement and rotational degrees of freedom at each node.

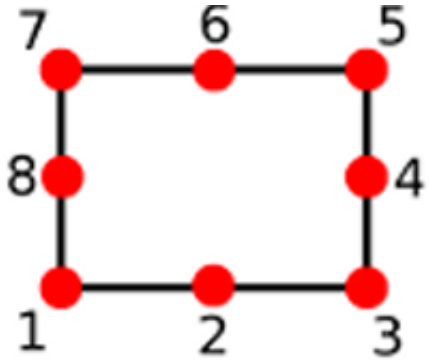
With first order shape function:





With second order shape function:



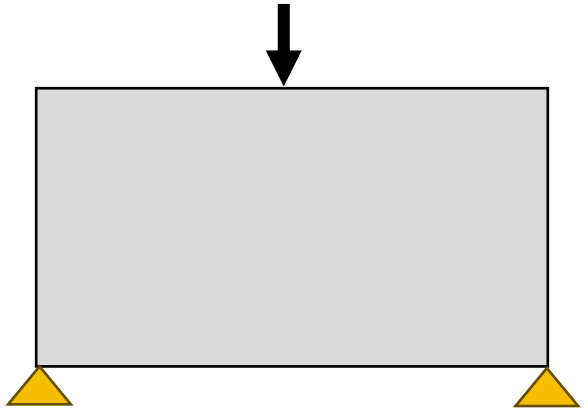


[6] Duffour, P (2026).

Element types

Can shell elements be used to model:

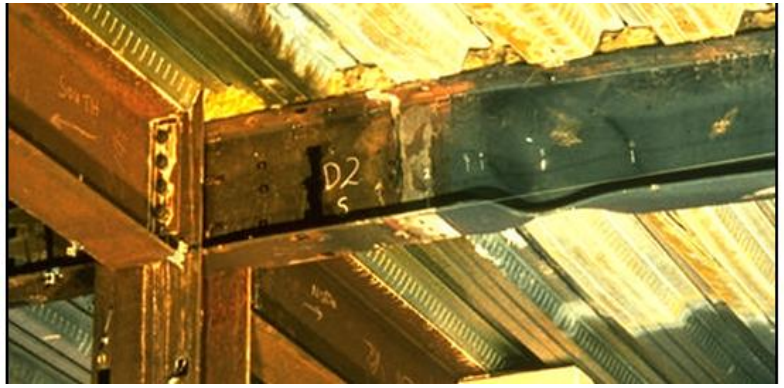
- Thick short beams?



- The thermal induced buckling of railway tracks?

[7] Mandal and Lees (2016)

- The thermal induced local buckling

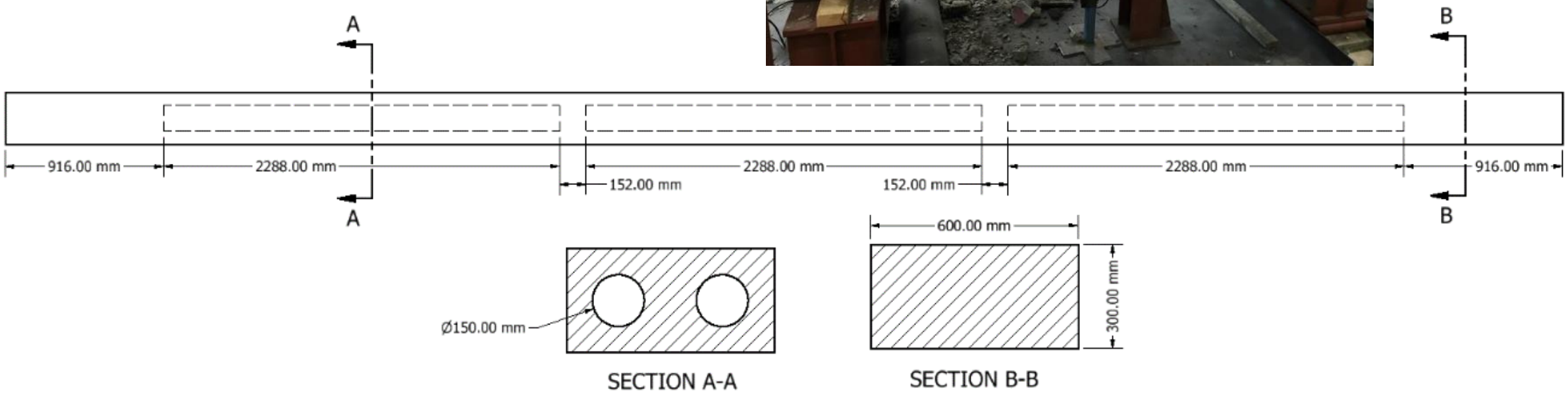


[4] Newman et al. (2000)

Element types

Can shell elements be used to model:

- Failure of a concrete girder?



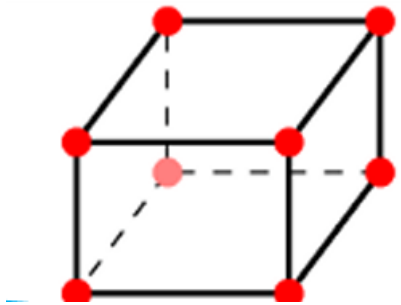
Element types

<u>3D element</u>

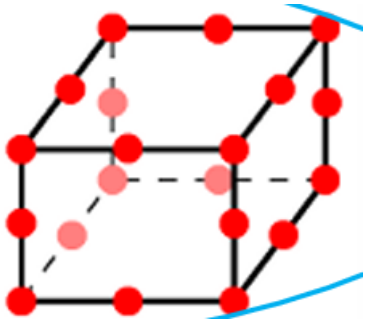
- No geometric reduction; field variable varies in three dimensions.

- Translation degrees of freedom at all nodes.

With first order shape function:



With second order shape function:

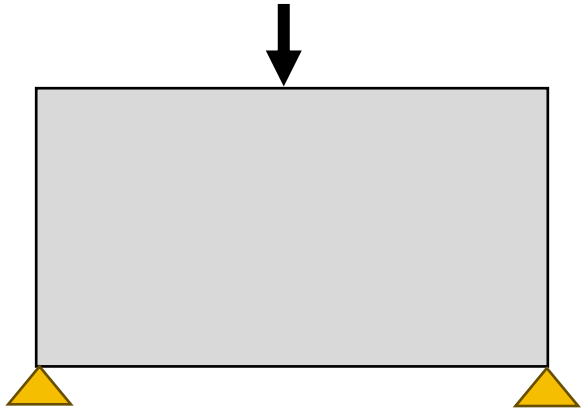


[6] Duffour, P (2026).

Element types

Can solid elements be used to model:

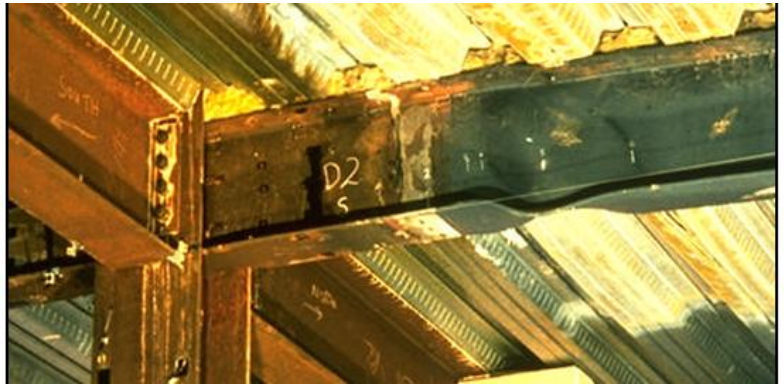
- Thick short beams?



- The thermal induced buckling of railway tracks?

[7] Mandal and Lees (2016).

- The thermal induced local buckling



[4] Newman et al. (2000)

Summary

Brief introduction to FE modelling and understand the effect of the following **fundamental aspects** for applying an FE model:

- Discretisation
- Shape functions
- Element types

Modelling principles

1. Use the simplest practical model
2. Estimate results before you run the analysis
3. Increment the complexity
4. When you get a result, assume that they may have errors
5. Troubleshoot errors

Modelling principles

1. Use the simplest practical model

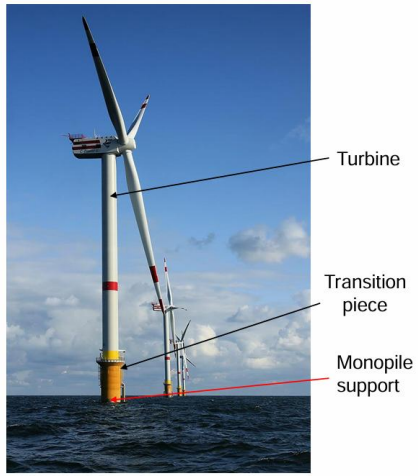
Define the modelling objectives:

- What questions are you trying to answer with the model?
- What accuracy do you require?
- What behaviour is important (global vs local)?

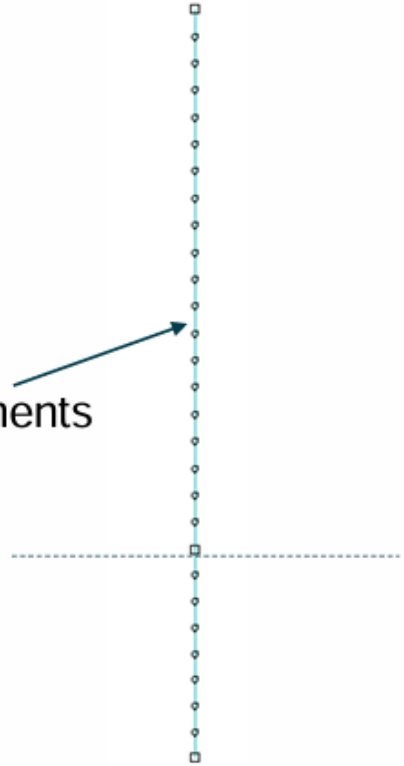
[6] Duffour, P (2026).

Modelling principles

[6] Duffour, P (2026)



Wire model of the whole system



1D elements

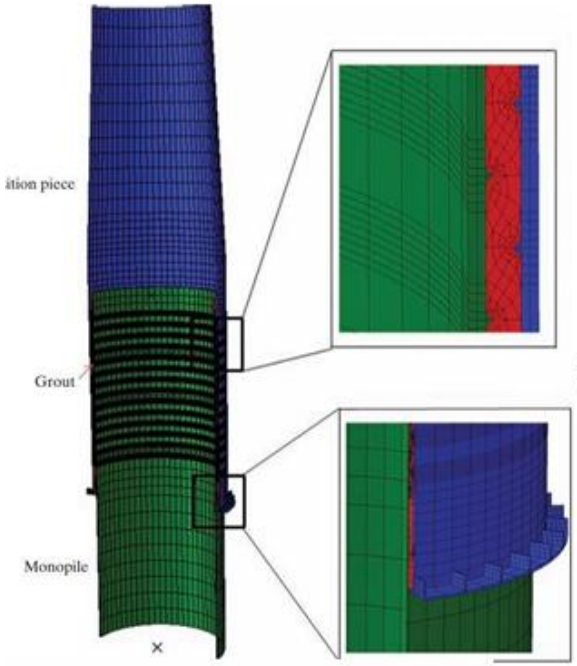
Gives overall dynamics behaviour and forces

Shell model of the monopile

2D elements

Used to check local buckling of cylinder

Detailed model of the transition piece



3D elements

Used to size steel details (stiffeners etc)

References

[1] MacLeod, I. A. (2005). *Modern structural analysis : modelling process and guidance / Iain A. MacLeod.* Thomas Telford

[2] Debney, P., & Institution of Structural Engineers. (2020). *Computational engineering / Peter Debney.* Institution of Structural Engineers.

[3] Khennane, A. (2013). *Introduction to finite element analysis using MATLAB and Abaqus / Amar Khennane.* Taylor & Francis.

[4] Newman, G. M., J. T. Robinson, and C. G. Bailey. (2000). *Fire Safe Design: A New Approach to Multi-Storey Steel-Framed Buildings*. SCI Publication Fire and Steel Construction 288. Steel Construction Institute.

[5] Sharma, K. (2019). *Introduction to FEA* [SlideShare slides]. SlideShare. https://www.slideshare.net/KishanSharma26/introduction-to-fea-77358694

[6] Duffour, P. Personal communications – CEGE0009 lecture slides, (2026).

[7] Mandal, N. K., & Lees, M. (2016). *An investigation into monitoring rail stress in continuously welded rails through stress-free temperature* [Conference paper]. Conference on Railway Engineering, Melbourne, Australia.